Final Experimental Report for Preliminary Screening: YOLO-Based Dental Caries Detection Implementation

Github Repositories: https://github.com/hejingxi6/Caries-Detection-Implementation

Executive Summary

An end-to-end YOLO-based dental caries detection pipeline was implemented for the preliminary screening task and then improved through three controlled optimization rounds. The main contribution of this work is a traceable engineering process: starting from a basic baseline, the detector was strengthened through threshold sweeping, imbalance-oriented training changes, and literature-guided design decisions.

The strongest retained operating point in the available evidence is Round 2 sweep for mAP@0.5 (0.595) and recall (0.592), while Round 3 sweep gives the best mAP@0.5:0.95 (0.425) and the highest precision (0.677). Compared with the initial baseline (mAP@0.5 = 0.312, mAP@0.5:0.95 = 0.216), this means the project achieved clear measurable improvement rather than only narrative progress.

1. Task and Reporting Basis

The preliminary-screening task was to complete a basic implementation of deep-learning-based dental caries detection within a short timeline and prepare for a follow-up technical discussion. This report therefore focuses on the completed pipeline, the retained experimental results from each round, and the technical interpretation of those results.

2. Experimental Design Across Rounds

To keep the comparison meaningful, the project followed one consistent evaluation logic: the same validation split was reused across rounds, final model quality was read from retained validation outputs, and each round also included a threshold sweep to separate raw training result from the selected deployable operating point. This is important because threshold choice, input size, and class imbalance can materially change detector performance even when the model family remains the same.

Round	Run name / evidence	Main change	Time cost	Reason for trying it
Initial baseline	baseline run	Basic YOLOv8 sanity pipeline	reference stage	Establish the initial YOLOv8 detector for later comparison
Round 1	fast_stage1_fix	Stabilized first full training run + threshold sweep	40 epochs; fast first comparison	Establish a reproducible first detector and identify the best validation threshold
Round 2	rebalance_imgsz768	Rebalanced training list / minority-class exposure + 768 input + sweep	51 epochs completed	Address clear class imbalance and recover more lesion recall
Round 3	lesionmix_imgsz960_e80	Lesion-mix style augmented training set + higher resolution (960) + sweep	69 epochs completed	Push lesion visibility and class exposure further while improving localization quality

Round 3 did not outperform the earlier rounds on every metric. Instead, it traded some recall for stronger precision and the best mAP@0.5:0.95. For that reason, the analysis separates coarse detection strength (mAP@0.5) from stricter localization quality (mAP@0.5:0.95), and interprets the selected operating point in relation to the screening objective.

3. Quantitative Results

Table 1 reports the strongest retained metrics from the available experimental evidence. The comparison is deliberately split into training result and sweep-best result because deployment quality depends on the operating threshold rather than on the raw training default alone. Here, mAP@0.5 means mean Average Precision at an IoU threshold of 0.5, while mAP@0.5:0.95 refers to the average AP computed across IoU thresholds from 0.5 to 0.95.

Stage	Variant	Precision	Recall	mAP@0.5	mAP@0.5:0.95	Notes
Initial baseline	baseline	0.331	0.362	0.312	0.216	Initial retained baseline

						result
Round 1	train	0.491	0.533	0.505	0.343	First stable full training result
Round 1	sweep-best	0.653	0.470	0.562	0.402	Best row: conf 0.25, IoU 0.50
Round 2	train	0.570	0.589	0.540	0.358	Rebalanced training clearly lifted recall
Round 2	sweep-best	0.590	0.592	0.595	0.403	Best row: conf 0.15, IoU 0.50
Round 3	train	0.613	0.545	0.523	0.365	Higher precision, slightly lower mAP@0.5
Round 3	sweep-best	0.677	0.491	0.577	0.425	Best row: conf 0.30, IoU 0.50

Compared with the initial baseline, the best available mAP@0.5 rose from 0.312 to 0.595 in the Round 2 sweep, while the best available mAP@0.5:0.95 rose from 0.216 to 0.425 in the Round 3 sweep. In practical terms, the project achieved clear improvement in both coarse detection and stricter localization, although the minority lesion classes remain challenging.

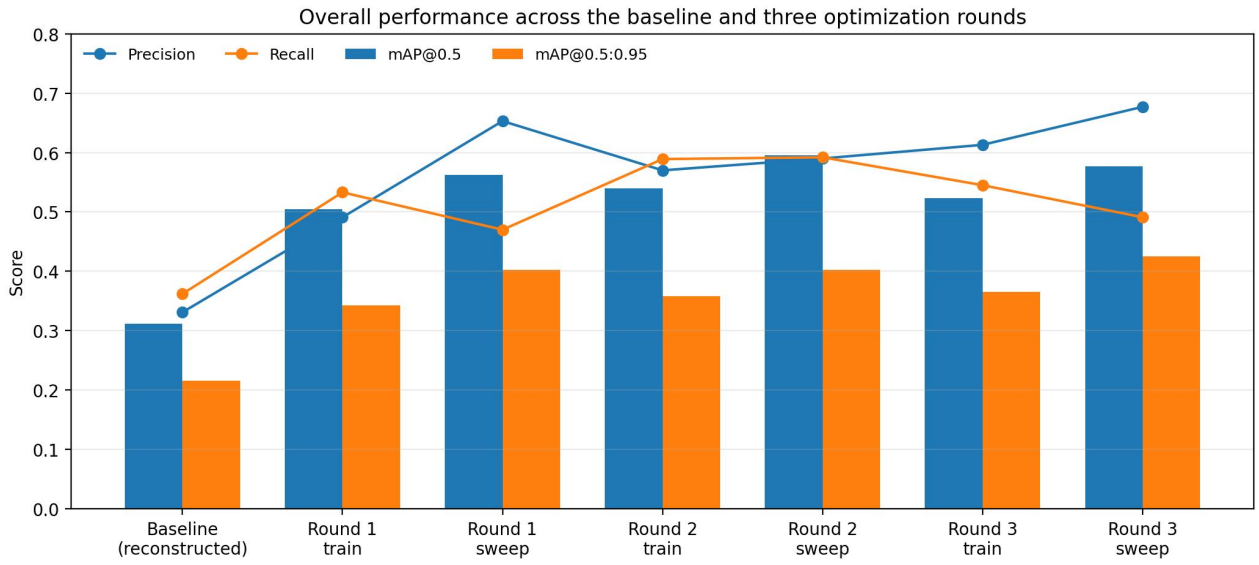


Figure 1. Overall performance across the baseline and the three optimization rounds.

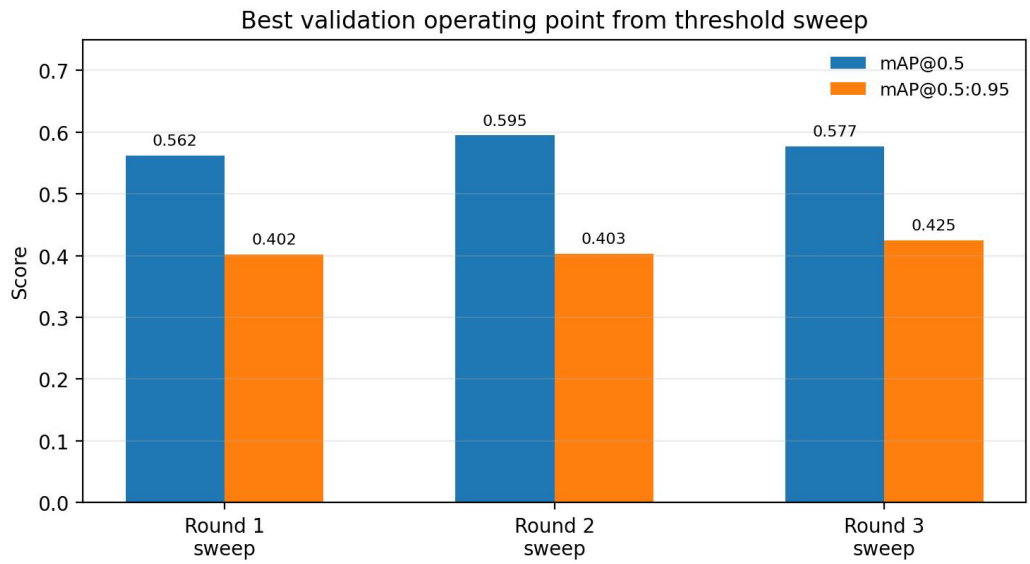


Figure 2. Best validation operating point selected by threshold sweep in each completed round.

4. Class-Wise Analysis

The model family consistently detects the Tooth class far more reliably than lesion classes. This is not surprising: Tooth has the largest number of visible instances, the clearest visual structure, and the most stable geometry. The three lesion classes are harder for different reasons: Caries is small and easily absorbed into background; Cavity is visually variable; Crack has extremely few examples and therefore unstable learning.

Class	Round 1 mAP@0.5	Round 2 mAP@0.5	Round 3 mAP@0.5	Interpretation
Caries	0.322	0.342	0.345	Only modest improvement; still frequently confused with background and tooth-

				like patterns
Cavity	0.429	0.412	0.397	Reasonable but unstable; gains from rebalancing were not preserved under the later high-resolution lesion-mix setup
Crack	0.299	0.437	0.384	Strong jump in Round 2 suggests rebalancing mattered; later decline reflects tiny sample size and instability
Tooth	0.976	0.970	0.966	Already near ceiling; this class is not the bottleneck

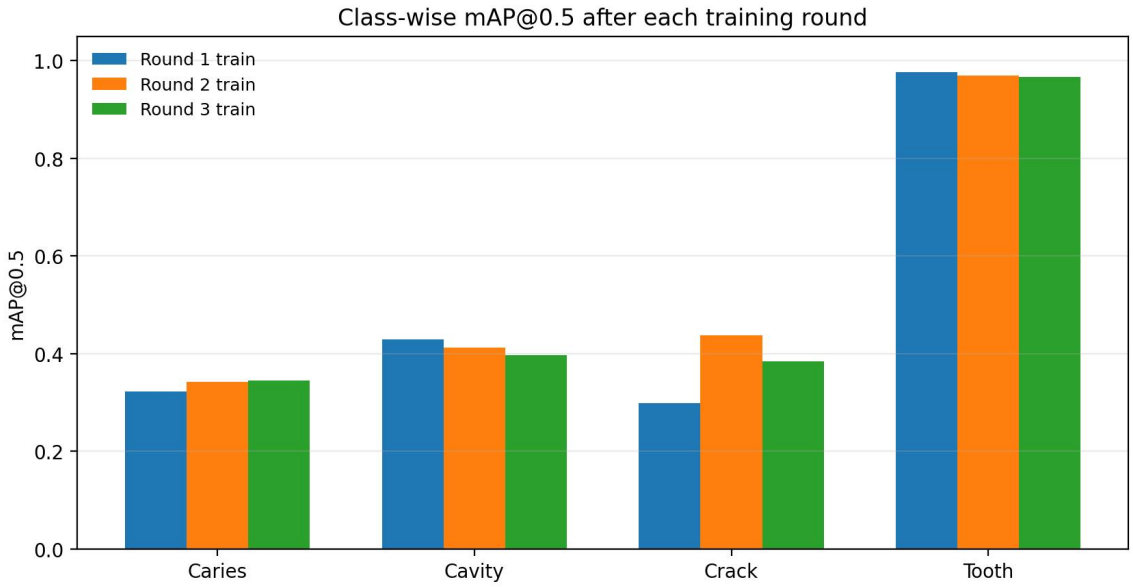


Figure 3. Class-wise mAP@0.5 across the three completed training rounds.

A report that only presents global mAP can look stronger than it really is, because clinically important failure modes may remain hidden. The class-wise results show the actual bottleneck in this project: tooth detection is already strong, while small lesion detection under class imbalance is still the main difficulty.

5. Error Analysis and Qualitative Review

The retained confusion matrix and validation-prediction panels show the same pattern as the metric table: many lesion instances are still lost to background, while tooth detection stays dominant. This means the model is not merely “low scoring”; it is systematically biased toward visually stronger and more frequent structures.

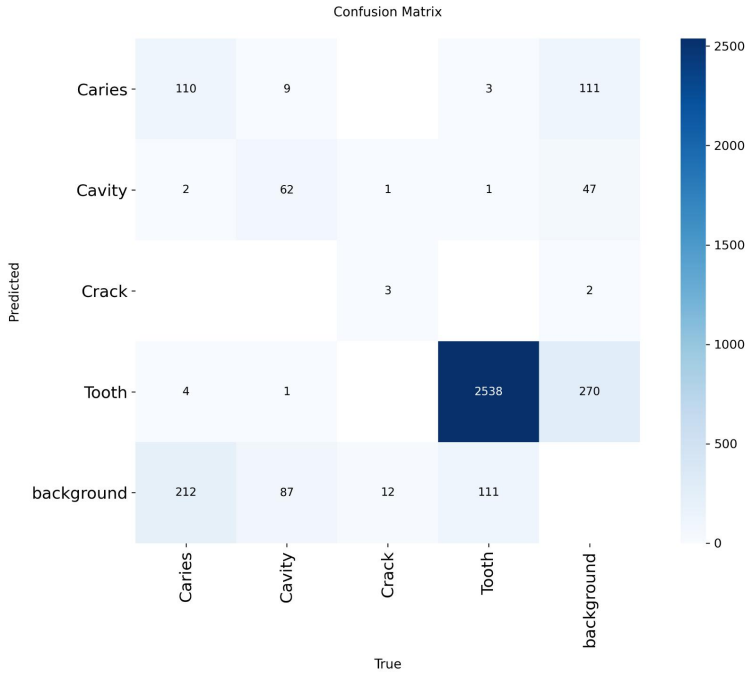
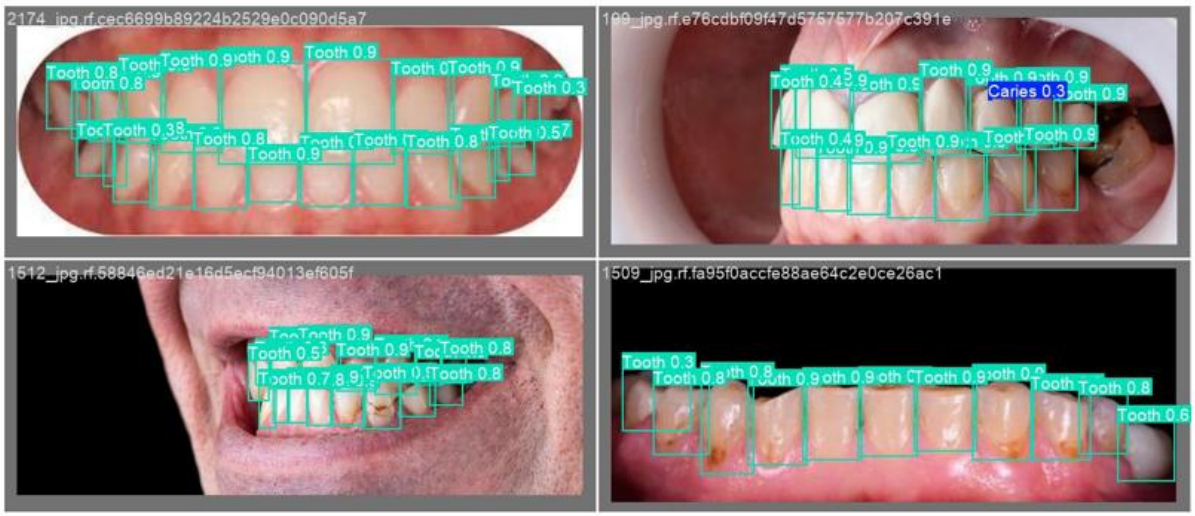
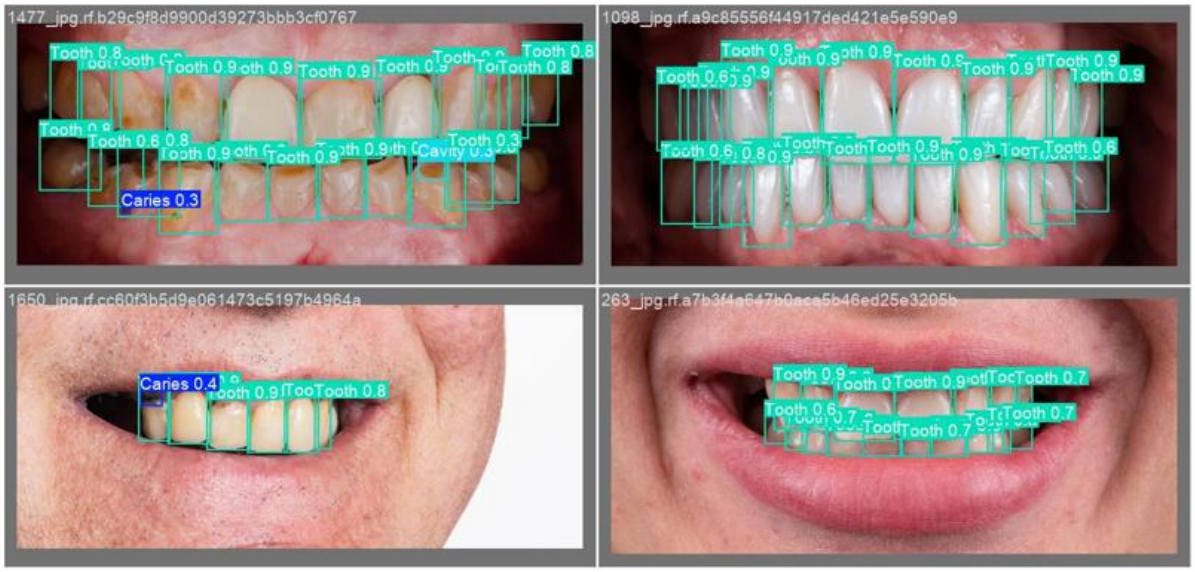


Figure 4. Retained confusion matrix export. Large background confusion remains for Caries, Cavity, and Crack.

Validation prediction panel 1



Validation prediction panel 2



Validation prediction panel 3

Figure 5. Representative validation prediction panels retained from the experiment artifacts.

These qualitative examples show both progress and remaining weaknesses. A positive signal is that lesion boxes do appear on genuinely difficult images, which indicates that the detector is learning more than a trivial ‘‘tooth = object’’ pattern. However, lesion recall remains sensitive to threshold choice, and several images still show dense tooth detections around sparse lesion evidence. Taken together, the preserved qualitative outputs are sufficient to illustrate the dominant validation pattern observed in this project: strong tooth detection, but still unstable minority-lesion detection under difficult visual conditions.

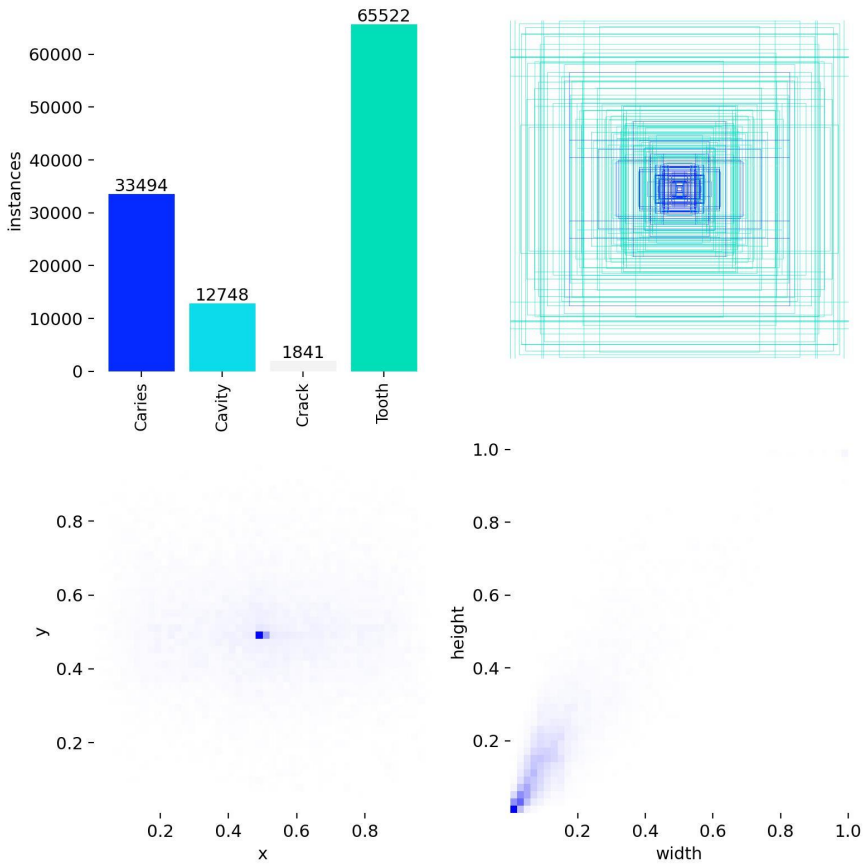


Figure 6. Retained label-distribution export from the lesion-mix round, showing that Tooth still dominates the training signal.

6. Literature-Guided Interpretation

Dental object-detection papers can report much stronger headline numbers than this project, yet they often use different modalities, curated institutional datasets, different label definitions, and different evaluation protocols. Therefore, those numbers should be used as methodological guidance rather than as a naive target for direct comparison.

Literature-backed finding	Evidence	How it maps to this project	Decision taken here
Threshold tuning materially changes dental detector	Dental YOLOv7 bitewing study [3]	Separating raw training metrics from sweep-best metrics is	Implemented in all three rounds

behavior		necessary	
Strong papers are often not directly comparable because datasets and annotation protocols differ	Bayati 2025, Ayhan 2025, AI-Dentify 2024 [1][2][4]	Do not pretend that a 0.59 mAP on DentalAI should match curated bitewing studies	Explained explicitly in this report
Small lesions benefit from slicing/tiling and high-resolution treatment	SAHI and small-object literature [6]	Supports the move toward larger images and future sliced inference	Partially implemented via higher-resolution Round 3; SAHI identified as next strong candidate
Long-tail imbalance must be addressed directly, not only by changing backbones	Focal Loss / Class-Balanced Loss [7][8]	Supports rebalancing and lesion-mix style augmentation	Implemented in Round 2 and Round 3
Annotation inconsistency can cap performance on small lesions	Kunt 2023 and AI-Dentify 2024 [4][5]	Explains why minority-class metrics remain unstable even after optimization	Discussed as a genuine limitation, not hidden

The literature is used here to interpret model behavior, separate transferable methods from non-comparable headline results, and keep the final judgment aligned with the actual experimental setting used in this project.

7. Overall Assessment, Limitations, and Next-Step Recommendation

Overall, the preliminary screening demonstrates a completed YOLO-based dental caries detection implementation together with a structured optimization process. A working detector was built, trained, validated, inspected, and iteratively improved across three rounds. The retained evidence shows clear engineering progress: compared with the initial baseline, later rounds achieved substantially better validation performance, while the literature review helped interpret the results in a methodologically grounded way.

However, the detector is not yet clinically reliable. The main limitation is that minority lesion classes remain unstable and are still heavily affected by background confusion and small-object difficulty. In addition, some early raw artifacts were not preserved, so the final judgment in this report is based on retained evidence only. This does not invalidate the observed improvement, but it does limit the completeness of retrospective comparison across all earlier experiments.

Among the completed rounds, Round 2 provided the strongest mAP@0.5 and recall after rebalancing, while Round 3 provided the strongest precision and mAP@0.5:0.95 after lesion-mix augmentation and higher-resolution training. Therefore, the most reasonable short-cycle next step is to test SAHI sliced inference on the strongest retained checkpoint while keeping the same validation split. This directly targets the small-lesion detection problem observed in the retained error patterns. In parallel, a small annotation-quality audit or relabel check for the hardest lesion images would also be valuable before attempting more complex backbone or training changes.

8. References

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